

Johnathon Beuter

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Education

Iowa State University of Science and Technology, Ames IA

- Master of Science in Cyber Security GPA: 3.77 January 2025 – May 2026
- Bachelor of Science in Cyber Security Engineering GPA: 3.21 August 2021 – May 2025
- Minor in Spanish

SANS Institute, Bethesda MD

- GIAC GFACT Certification December 2023 – May 2024

Professional Experience

Cyber Security Teaching Assistant

Iowa State University of Science and Technology, Ames IA | August 2023 – May 2026

- Responsible for troubleshooting virtual machine networking and infrastructure issues across 30+ labs serving 100+ students
- Revised 12+ lab exercises on lin/winPEAS, Bash scripting, nmap, and reverse shells, updating content and improving instructional quality
- Authored new course content focusing on lateral movement using SSH tunnels and Proxychains

Penetration Tester

IP Pathways, Urbandale IA | January 2026 – May 2026

- Led testing of public facing infrastructure and internal networks for numerous regional credit unions, charities, and school districts
- Solo-authored penetration test reports including executive summaries, risk ratings, proof-of-concept exploits, and remediation guidance for both technical and non-technical audiences
- Utilized Nessus, nmap scripting engine, Metasploit to identify and exploit vulnerabilities

Cyber Security Engineering Intern

Wellmark Blue Cross and Blue Shield, Des Moines IA | May 2025 – August 2025

- Built an isolated sandbox environment using CAPEv2, Python, and Virtual Box to automate malware analysis process
- Identified threats to Wellmark's use of Microsoft Configuration Manager (SCCM) and created a pen test plan using SharpSCCM and ntlmrelayx to verify vulnerabilities
- Detected and mitigated network threats daily, using Palo Alto firewalls and Windows Defender EDR

Red Team Intern

United States Air Force Research Laboratory, Rome NY | May 2024 - August 2024

- Engineered 3 offensive security tools using Nim, Python, and GO capable of performing Modbus denial of services attacks against Raspberry Pis simulating Operational Technology (OT) devices
- Led weekly offensive operations against Windows and Linux environments, leveraging custom exploit tooling and Sliver C2 to conduct lateral movement, establish persistence, and evade detection

- Reverse-engineered packed malware samples using Ghidra and x64dbg, producing detailed technical documentation of findings

Activities

Engineers Without Borders (EWB) Volunteer

- State of Iowa President: Implemented organization improvement informed by collaboration with other EWB chapters and industry professionals
- Vice President of Engineering: Led international team of volunteers in Ghana to install a solar borehole, build a pit latrine, and assist in community outreach initiatives

Hacking and Cybersecurity Club

- CTF Team Captain: Organized weekly participation of club in national and international CTF competitions, increasing club involvement in CTFs
- Led teams in 6 Cyber Defense Competition (CDC), averaging a top three finish including first place finishes in both the national and international Iowa State CDCs

Side Projects

Lateral Movement Lab [Fall 2025] - Personal

- Used Docker to make an isolated network environment for SSH tunneling and lateral movement, experimenting with Chisel to understand tunneling methods and corresponding detection techniques

Active Directory Lab [Spring - Fall 2025] - Personal

- Deployed an AD lab in VirtualBox to deepen hands-on network security administration experience, practice offensive techniques, and better understand Windows environments

Remote Access Trojan [Spring 2024] - Academic

- Built a C code-based RAT for a digital forensics course and performed forensic analysis with Ghidra, Process Monitor, and Windows Event Viewer to identify malware artifacts

MQTT Server [Fall 2023] - Personal

- Configured an Ubuntu server to run MQTT and engineered networking to support communication with multiple Raspberry Pis locally and across the internet

Awards and Accolades

Member of the Year 2025 - Iowa State EWB

- Recognized for outstanding membership in the Iowa State Engineers Without Borders chapter, contributing to project planning, overseas work, and community outreach initiatives that advanced organizational goals

ACE Class Valedictorian and Distinguished Graduate

- Graduated as Valedictorian of a 30-person cyber warfare training program (ACE), comprised of college students, military personnel, and civilian contractors from government agencies and top universities